

Chirag Veerwani

Sophomore at Indian Institute of Technology Hyderabad, India

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Education

Indian Institute of Technology Hyderabad (IITH)

B.Tech in Industrial Chemistry, Sophomore

Cumulative GPA: **9.00**

Intended Research Direction

Building foundations in computational mathematics, number theory, combinatorics, and algorithms, with an interest in symbolic computation and computer-assisted approaches to mathematics.

I am particularly interested in experimental and computational methods for exploring mathematical structures, and in how automation and formal tools can support mathematical reasoning and verification. I am also interested in applying mathematical and algorithmic ideas to scientific domains.

Research & Technical Experience

Undergraduate Research Assistant – Computational Chemistry

- Working with a faculty-led research group on computational modelling and algorithmic approaches to molecular and biochemical systems.
- Collaborating with PhD students to develop and test simulations and computational experiments relevant to biological and chemical processes.
- Contributing to code development, data analysis, and interpretation for ongoing research projects.

SageMath – Computational Mathematics (Independent Study)

- Studying SageMath as a tool for computational mathematics and symbolic computation.
 - Exploring number-theoretic and algorithmic experiments using Sage.
 - Reviewing open issues and project discussions to understand large mathematical codebases.
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Algorithmic Problem Solving

- Participating in Codeforces contests and practice to gain exposure to algorithmic problem solving and competitive programming.
 - Solving selected problems on Project Euler and Brainstellar to build mathematical intuition and computational thinking.
 - Studying foundational data structures and algorithms, including arrays, basic trees, graphs, introductory dynamic programming, and time complexity.
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Programming & Tools

- **Python:** basic working experience through multiple small projects and scripts; comfortable with control flow, functions, and modular code.
- Experience with scripting, simple automation, and logic-based programs.

- Exposure to data handling and visualization using Pandas, NumPy, and Matplotlib.
 - Introductory experience with web scraping and browser automation (Requests, BeautifulSoup, Selenium).
 - **C / C++**: basic familiarity, primarily through algorithmic problem solving and coursework.
 - **SageMath**: beginner-level use for computational mathematics and mathematical experimentation.
 - Development environments: PyCharm, Jupyter Notebook, Google Colab.
 - Version control: Git.
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Open Source & Competitions

Hacktoberfest (2025)

- Contributed **6 merged pull requests** across open-source repositories.
- Among the first **10,000 contributors** to complete Hacktoberfest requirements.

International Math Optimization Challenge (IIMOC)

- Awarded Silver Certificate (Team Event).
 - Participated in algorithmic problem solving and implementation.
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Leadership & Activities

- Head, Dramatics and Theatre Arts Club, IITH
 - Active Member, Web & Technical Team, International Relations Cell
 - PR coordinator, College Cultural Festival
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